5 CBS, USSD, MMI Data Coding Scheme

((This subsection should really be moved to the CBS, USSD, MMI standards: 3GPP TS 23.041 Technical realization of the Cell Broadcast Service (CBS), 3GPP TS 23.090 Unstructured Supplementary Service Data (USSD) - Stage 2 or TS 24.090 Unstructured Supplementary Service Data (USSD) - Stage 3, 3GPP TS 22.030 Man-Machine Interface (MMI) of the User Equipment (UE).))

Bits 7..4	***Use of bits 3..0 (sometimes 5..0)***
((Neither of these ways of language tagging make much sense; deprecate?)) Otherwise: say "General Data coding indication with language indication", "Uncompressed" (of course) and "no message class" (and 8-bit is not covered).	
0000 (0x00..0x0F) **7-bit alphabet as set in the TP User Data Header, language indication for alphabets 0x00, 0x10, or 0x11**	**For other alphabets: 0000..1110 are reserved, and group 0000 should not be used.** 0000 (0x00) German 0001 (0x01) English 0010 (0x02) Italian 0011 (0x03) French 0100 (0x04) Spanish 0101 (0x05) Dutch 0110 (0x06) Swedish 0111 (0x07) Danish 1000 (0x08) Portuguese 1001 (0x09) Finnish 1010 (0x0A) Norwegian 1011 (0x0B) Greek if alphabet is 0x00 (deprecated); though may be used for alphabet 0x11. 1100 (0x0C) Turkish 1101 (0x0D) Hungarian 1110 (0x0E) Polish 1111 (0x0F) Language is unspecified. Common use. Shall be used for 7-bit alphabets other than 0x00, 0x10, 0x11. Continued at group 0010 for alphabet 0x10.

0001 (0x10..0x1F): **Language indication in a message prefix**	**7-bit alphabet as set in the TP User Data Header** 0000 (0x10) Message preceded by language tag. The first 3 characters of the message are a two-character representation of the language encoded according to ISO 639-1 [12], followed by a CR character. The CR character is then followed by (up to 90) 7-bit code units of text. If the alphabet is 0x00 (deprecated) and the language tag is “el<CR>”, then use the Greek special base table for alphabet 0x00, 6.2.1.3, deprecated. (NOTE 1). An MS not supporting the set 7-bit alphabet **may** present the two character language identifier followed by improperly interpreted user data. NOTE 2: Only two-letter language indications can be used. NOTE 3: The language tag prefix is in the alphabet 0x00, and consists of lowercase letters a-z.
	UCS2/UTF-16BE encoding 0001 (0x11) The character coding is UCS2/UTF-16BE (except for the language tag) and the message is preceded bya two-letter language tag. The full message starts with two 7-bit alphabet (0x00) character representation of the language code encoded according to ISO 639-1 [12]. This is padded to the octet boundary with two bits set to 0 and then followed by (up to 40) 16-bit code units of UTF16BE-encoded message. (NOTE 1). An MS not supporting the UCS2/UTF16BE coding **may** present the two character language identifier followed by improperly interpreted user data. NOTE 2: Only two-letter language indications can be used. NOTE 3: The language tag prefix is in the 7-bit alphabet 0x00, and consists of lowercase letters a-z, but the message is in UCS2/UTF-16BE.
	0010..1111 (0x12..0x1F) Reserved
0010 (0x20..0x2F) **7-bit alphabet as set in the TP User Data Header, Language language indication for alphabet 0x10 (continued from group 0000)**	**For other 7-bit alphabets: 0000..1111 are reserved, and group 0010 should not be used.**
	0000 (0x20) Czech
	0001..0011 (0x21..0x23) Reserved
	0100 (0x24) Icelandic
	0101 (0x25) Romanian
	0110 (0x26) Albanian
	0111 (0x27) Catalan
	1000 (0x28) Croatian
	1001 (0x29) Slovak
	1010 (0x2A) Estonian
	1011 (0x2B) Latvian

	1100 (0x2C) Lithuanian		
	1101 (0x2D) Vietnamese		
	1110 (0x2E) Irish		
	1111 (0x2F) Tagalog		
0011 (0x30..0x3F)	0000..1111 (0x30..0x3F) Reserved for other languages when using a 7-bit alphabet in future versions of this standard		
01xx (0x40 .. 0x7F)	**General Data Coding indication**		
	Bit 5, if set to 0, indicates the text is uncompressed		
	*Bit 5, if set to 1, indicates the text is compressed using the compression algorithm defined in 3GPP TS 23.042 [13] (**strongly deprecated;** 3GPP TS 23.042: **to withdraw**)*		
	Bit 4, if set to 0, indicates that bits 1 to 0 are reserved and have no message class meaning; interpret as Class 1.		
	Bit 4, if set to 1, indicates that bits 1 to 0 have a message class meaning.		
	Bit 3	Bit 2	Alphabet:
	0	0	**The 7-bit alphabet as set in the TP User Data Header (default alphabet ID is 0x00)**
	0	1	8 bit data
	1	0	UCS2/**UTF-16BE** (16 bit) [10]
	1	1	Reserved
	Bit 1	Bit 0	Message Class (if bit 4 is set to 1):
	0	0	Class 0: Immediate.
	0	1	Class 1: Default meaning: ME-specific.
	1	0	Class 2: (U)SIM specific message.
	1	1	Class 3: Default meaning: TE-specific (see 3GPP TS 27.005 [8])
1000 (0x80..0x8F)	Reserved coding group		
1001 (0x90..0x9F)	**Message with (secondary?) User Data Header (UDH) structure (prefix?)**		
	Bit 3	Bit 2	Alphabet:
	0	0	**The 7-bit alphabet as set in the TP User Data Header (default alphabet ID is 0x00)**
	0	1	8 bit data
	1	0	UCS2/**UTF-16BE** (16 bit) [10]
	1	1	Reserved
	Bit 1	Bit 0	Message Class:
	0	0	Class 0: Immediate.

	0	1	Class 1: Default meaning: ME-specific.
	1	0	Class 2: (U)SIM specific message.
	1	1	Class 3: Default meaning: TE-specific (see 3GPP TS 27.005 [8])
1010..1100 (0xA0..0xCF)	Reserved coding groups		
1101 (0xD0..0xDF)	I1 protocol message defined in 3GPP TS 24.294 [19] (USSD only)		
1110 (0xE0..0xEF)	~~Defined by the WAP Forum [15]~~ highly obsolete reserved		
1111 (0xF0..0xFF)	**?????????? the 1111 group seems to be an editorial mistake** reserved		
	Bit 3	Bit 2	Alphabet:
	0	0	**The 7-bit alphabet as set in the TP User Data Header (default alphabet ID is 0x00)**
	0	1	8 bit data
	1	0	Reserved (*UCS2/**UTF-16BE** (16 bit) [10]...)*
	1	1	Reserved
	Bit 1	Bit 0	Message Class:
	0	0	Class 0: Immediate.
	0	1	Class 1: Default meaning: ME-specific.
	1	0	Class 2: (U)SIM specific message.
	1	1	Class 3: Default meaning: TE-specific (see 3GPP TS 27.005 [8])